

Architecture Specification & Product Requirements Document (PRD)

Autonomous Agentic GTM Engine & Value-Hook Pipeline

Target System: ReactFirst AI Proactive Outbound Engine

Document Status: Ready for Implementation / Production-Grade Spec

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1. Motivation & Business Context

1.1 Problem Statement

In 2026, consumer-facing industries—specifically FMCG (Fast-Moving Consumer Goods), Direct-to-Consumer (DTC) fashion, and modern Retail—operate inside incredibly narrow narrative windows. Traditional cold outbound strategies (e.g., static filtering via LinkedIn Sales Navigator, generalized email sequences) yield sub-1% reply rates. This failure stems from a lack of immediate, context-specific value injected at the point of initial contact. Standard static code pipelines cannot adaptively crawl the web, identify qualitative social media vulnerabilities, cross-reference them against complex operational constraints, and programmatically generate bespoke analytical collateral (e.g., tailored Narrative Vulnerability PDFs) for every distinct brand in real time.

1.2 The Agentic Solution

This document specifies a production-grade, highly autonomous AI Agent that orchestrates an advanced Go-To-Market (GTM) data pipeline. By combining state-of-the-art Large Language Models (LLMs), a highly localized hybrid RAG mechanism, robust mathematical validation layers, and resilient execution loops, the agent autonomously transforms raw market intent into highly contextualized outbound campaigns.

Instead of waiting for an enterprise to experience a public relations crisis or a narrative shift passively, this system hunts for high-fit brands, analyzes their public digital footprints in controlled micro-batches (Chunks), grades their alignment against strict Ideal Customer Profile (ICP) parameters, pairs them with contextual historical case studies, and calls the core ReactFirst application API to produce a customized PDF value proposition ready for automated, multi-channel distribution.

2. Authoritative Data Input Architecture

The agent operates over three strictly bounded data inputs and an explicit GTM instruction set. The system must treat these sources as the single version of truth, preventing the model from hallucinating baseline configurations, brand realities, or pricing models from its pre-trained parametric knowledge base.

2.1 Brands Data Catalog (`brands_catalog.csv`)

This file contains the historical context of existing customers, past opportunities, and blacklisted entities. It consists of 9 immutable columns:

- `Uniq_Id`: Primary key (UUID string).
- `Brand_Name`: Legal and public trading name of the company.
- `Primary_Domain`: The authoritative root URL (used for web scraping and identity de-duplication).
- `Core_Category`: Multi-tier categorization path (e.g., `Apparel > Athleisure > Sustainable`).
- `Estimated_Ad_Spend_Tier`: Annualized digital ad spend matrix (`Tier 1: $5M+`, `Tier 2: $1M-$5M`, `Tier 3: <$1M`).
- `Current_Status`: Operational state within the ReactFirst ecosystem (`Active_Client`, `Open_Oppportunity`, `Unreached_Prospect`, `Blacklisted`).
- `Historical_Social_Incidents`: Integer counting past viral narrative crises or PR vulnerabilities tracked.
- `Main_Competitor_Id`: Foreign key pointing to the primary competitor's `Uniq_Id`.
- `Gtin_Prefix`: Global Trade Item Number block identifying their product lines.

2.2 CRM Contact Store (`contacts.json` queried via `lead_store.py`)

To mimic a live MongoDB instance while eliminating network overhead and container management during rapid deployment cycles, contact records are handled via `mongomock` in-memory structures initialized through an import-safe data loader:

Python

```
lead_store.py #
import
import

None

def get_lead_data_collection
global
if is None
'
'gtm_db
'contacts
with open 'contacts.json' 'r' as

return
```

Each record conforms to the following strict JSON schema structure:

JSON

```
"#schema" "http://json-schema.org/draft-07/schema$"
```

```

        "title" "ContactRecord"
        "type" "object"
        "properties"
        "first_name" "type" "string"
        "last_name" "type" "string"
        "email" "type" "string" "format" "email"
        "corporate_access_key" "type" "string"
        "role" "type" "string"
        "linkedin_url" "type" "string" "format" "uri"
        "interaction_history_count" "type" "integer"
        "opt_out_status" "type" "boolean"
        "target_brand_id" "type" "string"

required" "first_name" "last_name" "email" "corporate_access_key" "role" "
        ""target_brand_id

```

2.3 GTM Operational Policies (gtm_policies.txt)

An absolute instruction matrix parsed by the agent before finalizing any outbound execution state:

1. **Authoritative Context Bound:** Any claim regarding a prospect's market position, tier, or competitor layout must derive solely from `brands_catalog.csv`.
2. **ICP Validation Threshold:** A prospect brand is qualified for automated outreach if and only if it explicitly ticks 3\$ strict qualification parameters during deep scraping.
3. **The Premium Pricing Tier Tiering Loop:** Prospects belonging to companies with an `Estimated_Ad_Spend_Tier` of `Tier 1` qualify for the enterprise custom platform SLA. If their historical tracked incidents exceed 5, a premium risk multiplier of 15% is programmatically added to their internal value estimation calculation.
4. **Data Protection & Authentication Gate:** The `corporate_access_key` must be verified through the authentication tool before extracting or modifying private contact records or logging past interaction counts.
5. **Output Suggestions Ceiling:** When generating recommendations or matches, the agent must output no more than 3 distinct target angles or product capabilities. If a query requests a specific subset (e.g., "top 2 items"), the agent must output exactly that count.
6. **Explicit Zero-Match Boundary:** If no prospect matches the criteria or if the pipeline parameters fail validation, the agent must bypass all generative prose and return exactly the string: `"We have no product available today that fits your request"`.

3. Multi-Modal Agentic Core & Tool Topology

The architecture separates the discovery phase from the analytical execution loop. It relies on explicit tool schemas specifying detailed input boundaries to eliminate token wastage and prevent the agent from getting stuck in iterative execution loops.

3.1 Phase 1 Tooling: The Adaptive Discovery & 3-Way Fan-Out

Tool 1: `generate_search_queries`

- **Execution Environment:** *Gemini Flash Latest*
- **Purpose:** Generates a matrix of 10 to 20 highly granular, non-overlapping search query variations based on an amorphous vertical input descriptor.
- **JSON Schema Spec:**

JSON

```
    "name" "generate_search_queries"
    "description" "Transforms high-level market vertical definitions into 10-20 distinct, targeted"
    ".search queries to capture niche industry footprints. Use when initial seed datasets are low"
    "parameters"
    "type" "object"
    "properties"
    "vertical_seed" "type" "string" "description" "The base sector, e.g., 'sustainable footwear'"
    "Europe"
    "target_count" "type" "integer" "default" 15"
    "required" "vertical_seed"
```

Tool 2: `execute_3way_fanout`

- **Execution Environment:** Concurrent multi-model execution pool.
- **Purpose:** Executes the search query array simultaneously across three distinct discovery vectors to guarantee deep market scraping containment.
- **Mechanics:**
 1. *Vector A (Gemini Discovery):* Calls `gemini-3.1 flash-lite` with web grounding enabled to extract lists, press releases, and structured industry indices.
 2. *Vector B (SerpAPI + Maps):* Pulls raw Google Search and localized Maps node arrays to identify local physical retail presence and core domain references.
 3. *Vector C (Tavily Recovery):* Acts as a dynamic fallback engine. If Vector A and Vector B yield fewer than 2 structural domains for a given query line, Tavily is triggered automatically with specialized research switches enabled to uncover niche DTC brands.
- **JSON Schema Spec:**

JSON

```
    "name" "execute_3way_fanout"
    "description" "Executes concurrent parallel queries across Gemini Discovery, SerpAPI, and"
    ".Tavily Recovery fallback paths to maximize target yield"
    "parameters"
    "type" "object"
    "properties"
```

```

    "queries"
    "type" "array"
    "items" "type" "string"
    ".description" "List of validated search query strings generated by the Query Generator"

```

```

    "required" "queries"

```

Tool 3: `extract_and_score_pool`

- **Execution Environment:** *gpt-4o-mini*
- **Purpose:** Compiles all discovery raw data arrays, eliminates domain duplicates (De-duplication layer), and maps them against raw historical vectors from `brands_catalog.csv` to output a clean list of candidate companies.

3.2 Phase 2 Tooling: The Durable Enrichment Swarm & Core PDF Generation

To protect enterprise resources from thread exhaustion and rate limits, the agent processes candidates in deterministic micro-batches: **100 leads per chunk**, allocating a strict time budget of **800 seconds per chunk execution step**.

Tool 4: `analyze_company_chunk`

- **Execution Environment:** *gpt-5-mini* + Firecrawl API
- **Purpose:** Crawls the targets' root domains to extract tech stack metadata (detecting if the TikTok Pixel, Meta Pixel, or Google Tag Manager are present) and text patterns to determine their operational scale.
- **JSON Schema Spec:**

JSON

```

    "name" "analyze_company_chunk"
    "description" "Processes a controlled batch of exactly 100 domain entries. Crawls homepages"
    ".and legal footprints using Firecrawl to identify pixel configuration and operational data"
    "parameters"
    "type" "object"
    "properties"
    "domains"
    "type" "array"
    "items" "type" "string"
    ".description" "Array of exactly or up to 100 unverified target root domains"

    "required" "domains"

```

Tool 5: `evaluate_icp_tags`

- **Execution Environment:** Structural JSON Classifier Rule Engine.
- **Purpose:** Validates the extracted metadata against the qualifying criteria specified in Policy 2 (\$\ge 3\$ matching tags required for automatic qualification).

- **JSON Schema Spec:**

JSON

```

        "name" "evaluate_icp_tags"
description" "Applies a strict boolean filter across company profiles. Only companies meeting 3"
        ".or more defined ICP tags are qualified for programmatic PDF generation
        "parameters"
        "type" "object"
        "properties"
        "company_profile_data"
        "type" "string"
        ".description" "The raw text and technical string metadata extracted during the crawl step"

        "required" "company_profile_data"

```

Tool 6: match_solicitation_angle

- **Execution Environment:** Local Vector Index (ChromaDB + all-MiniLM-L6-v2) + Reciprocal Rank Fusion (RRF).
- **Purpose:** Executes a local hybrid RAG query combining semantic embeddings (representing the company's messaging vulnerabilities) with exact string matching (BM25) against ReactFirst's library of past historical crisis case studies. This process classifies the account into one of 4 specific outreach priority tiers (Tier 1: Critical Fit to Tier 4: No Match).
- **JSON Schema Spec:**

JSON

```

        "name" "match_solicitation_angle"
description" "Performs an RRF hybrid vector search against past enterprise case studies to"
        ".map out the exact outbound hook angle based on discovered public vulnerabilities
        "parameters"
        "type" "object"
        "properties"
        "scraped_narrative_context" "type" "string" "description" "The textual core messaging"
        ".extracted from the target company's site
        "category_path" "type" "string" "description" "Authoritative category matching from the"
        ".CSV database

        "required" "scraped_narrative_context" "category_path"

```

Tool 7: request_reactfirst_pdf

- **Execution Environment:** Programmatic System Runtime Client.
- **Purpose:** Executes a direct API call to the core ReactFirst application instance, triggering the generation of a specialized Narrative Analysis PDF report for the verified domain. It saves the resulting artifact directly to the workspace storage layer.

- JSON Schema Spec:

JSON

```

        "name" "request_reactfirst_pdf"
description" "Communicates directly with the ReactFirst product backend via secure API to"
        ".generate and store the value-hook analysis PDF asset
        "parameters"
        "type" "object"
        "properties"
        "target_domain" "type" "string"
        "validated_angle_key" "type" "string"
        "calculated_risk_score" "type" "number"

    "required" "target_domain" "validated_angle_key" "calculated_risk_score"

```

Tool 8: `secured_calculator`

- **Execution Environment:** Python AST Isolated Walk Engine.
- **Purpose:** Evaluates premium pricing increments and risk tier offsets (Policy 3) without introducing arbitrary code execution vulnerabilities. **Raw `eval()` or `exec()` loops are strictly prohibited.**
- **Implementation Standard:**

Python

```

import
import

:class SafeCalculator

    @classmethod
    def evaluate(
        cls,
        code: str
    ) -> str:
        return eval(code)

    @classmethod
    def _walk(
        cls,
        node: ast.AST
    ) -> str:
        if isinstance(node, ast.Call):
            return cls._walk(node.func)
        elif isinstance(node, ast.Str):
            return node.s
        else:
            return node.s

    @classmethod
    def type(
        cls,
        node: ast.AST
    ) -> str:
        if isinstance(node, ast.Call):
            return cls._walk(node.func)
        elif isinstance(node, ast.Str):
            return node.s
        else:
            return node.s

```



```

        return
    if
    in
elif isinstance
    type
    if
    in
    return
    "{raise
    f"Unauthorized mathematical syntax block: {node

```

4. Execution Flow Topology & Conversational Triggers

To illustrate the behavioral sequencing of the system, the following table charts how specific incoming business queries map directly to specific backend tools, ensuring complete rule compliance at every stage.

#	Incoming System Query Sample	Target Architectural Path & Execution Steps	Primary Tool Chain
Q1	"My access key is Access99 and my contact email is cmo@nordicwear.com. We have experienced 6 high-profile public incidents this year and our ad spend tier is Tier 1. What is our premium estimation tier, and is there a localized asset ready?"	1. Authenticate contact and cross-reference records via lead_store.py. 2. Read Policy 3 (Premium risk threshold calculation loop). 3. Compute current estimation pricing tier plus 15% risk increment using	authenticate $\rightarrow$ get_lead_data_collection $\rightarrow$ secured_calculator

		the isolated calculator. 4. Retrieve existing brand asset matches.	
Q2	"What are the structural profile differences between Brand_Id_01 (Apparel) and Brand_Id_02 (Footwear) in terms of digital ad footprints?"	1. Read catalog boundaries. 2. Fetch structural columns for both entries. 3. Generate a clean markdown table comparing categories, domains, and competitor metrics.	brands_catalog parsing via pandas

Q3	"Find apparel brands with an annual ad spend tier under Tier 2 that exhibit specific semantic weaknesses around ethical sourcing messaging."	1. Apply a hard column filter on the CSV data source. 2. Pass resulting domain frameworks to the local RAG vector space to surface companies displaying relevant thematic messaging gaps.	filter_products $\rightarrow$ match_solicitation_angle (Hybrid RAG)
Q4	"I want to track emerging cosmetic lines using sustainable packaging that don't have any current active Meta tracking codes running."	1. Execute the 3-Way Fan-Out engine to discover new domains. 2. Use the durable processing engine to check pixel setups in chunks of 100.	generate_search_queries $\rightarrow$ execute_3way_fanout $\rightarrow$ analyze_company_chunk

		3. Filter out domains with active tracking.	
Q5	"Give me a list of the top 5 brands most vulnerable to upcoming TikTok algorithm updates."	1. Identify relevant records. 2. Intercept output constraints using Policy 5. 3. Cap generative lists at exactly 3 results, overriding the user's initial request for 5.	Internal System Prompt Rule Containment
Q6	"We need to hunt down micro-influencer brands in the coffee niche that have no current products in our historical catalog database."	1. Generate search matrix. 2. Run 3-way scraping loop. 3. Check results against the authoritative	execute_3way_fanout \$\rightarrow\$ Return exact fallback: <i>"We have no product available today that fits your request"</i>

		CSV catalog. If none exist and scraping fails to meet the criteria, apply Policy 6.	
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5. Architectural Guardrails & Resiliency Patterns

To ensure this system operates safely and effectively within a lean engineering environment, it includes three foundational guardrails directly in its design.

5.1 Tool Gateway Validation Pattern

Every structural payload returned by the agent loop that targets external interaction infrastructures (e.g., passing lead objects to Smartlead or updating the production database workspace) must clear a strict gateway abstraction layer.

- This layer runs validation tests to ensure no fields contain null objects, strings match target formatting regexes perfectly, and generated PDFs exist in storage with valid headers before executing a step.

5.2 Trust-Gated Autonomy

Prospects that clear the structural `evaluate_icp_tags` processing layer with a borderline rating (e.g., companies that meet exactly 3 criteria but score low on traffic indicators) are barred from proceeding immediately to automated email delivery.

- Instead, the system routes these entries to a specific Slack channel via webhook, allowing a human growth operator to approve or discard the target with a single click. This creates a secure, human-in-the-loop framework for ambiguous data points.

5.3 Technical System Bounds & Anti-Loop Safeguards

- **Global Step Cap:** Any single query execution path entering `answer_question` is restricted to a maximum of 15 iterative tool calls. If the agent fails to reach a definitive final answer within 15 turns, it must stop processing and fall back to a safe error state.
- **Network Isolation Constraints:** Production outbound campaigns must use dedicated tracking sub-domains (`outreach.reactfirst.ai`) to keep cold outreach volume completely separated from the company's core corporate email domain infrastructure.
- **Rate and Volume Envelope:** Outbound email tools are throttled at a strict maximum of 50 messages per day per active outbound inbox.

6. Analytical Engineering Metrology (System Validation Formulas)

The success and operational quality of this agentic pipeline are measured using precise engineering metrics, avoiding vague vanity benchmarks like total email volume or raw click counts.

6.1 Pipeline Value Generation Velocity (\$V\$)

Measures the velocity at which the agent injects qualified, high-value opportunities into the CRM pipeline. This is heavily optimized by automating asset preparation and cutting down total cycle time:

$$\$V = \frac{N \times W \times A}{T} \$$$

Where:

- $\$N$ = Total number of verified ICP opportunities surfaced and processed by the agent.
- $\$W$ = Historic close rate percentage of value-hook outreach methodologies.
- $\$A$ = Average Annual Contract Value (ACV) of the ReactFirst enterprise subscription model.
- $\$T$ = Total sales cycle length measured in days.

6.2 Cost Per Validated Learning (\$C_{\text{VL}}\$)

Quantifies the operational cost efficiency of running automatic market discovery cycles across new verticals compared to hiring manual research agencies:

$$\$C_{\text{VL}} = \frac{C_{\text{infra}} + C_{\text{tokens}} + C_{\text{scraping_credits}}}{N_{\text{experiments}}} \$$$

Where:

- $\$C_{\text{infra}}$ = Baseline host computing costs.
- $\$C_{\text{tokens}}$ = Direct LLM context window input/output token costs (OpenAI + Gemini endpoints).
- $\$C_{\text{scraping_credits}}$ = Firecrawl and SerpAPI query usage expenditures.
- $\$N_{\text{experiments}}$ = Number of distinct sub-vertical market discovery loops successfully executed.

6.3 Signal-To-Campaign Latency (\$L_{\text{signal}}\$)

Tracks the responsiveness of the system by measuring the total time elapsed from initial discovery to delivery of the personalized value proposition:

$$\$L_{\text{signal}} = T_{\text{delivery}} - T_{\text{discovery}} \quad \text{left(Target Boundary Condition: } L_{\text{signal}} \leq 900 \text{ seconds } \text{right)} \$$$

7. Standard Operating Procedure (SOP) Checklist

Developers must verify all items on this checklist before pushing code changes to production:

- [] **Schema Conformance:** Verify that the payload structure returned by `get_lead_data_collection` maps directly to the `mongomock` storage layout without altering the original keys.
- [] **Isolated Calculation Testing:** Validate that passing complex mathematical expressions (e.g., `(1700 + 450) * 1.15`) to `SafeCalculator` evaluates correctly, and that any string containing characters outside the approved AST whitelist raises an immediate `ValueError`.
- [] **Strict String Fallback:** Confirm via automated integration tests that any query producing an empty result set yields exactly the string `"We have no product available today that fits your request"`, with no additional punctuation, spaces, or generative text.